# Arista Access Node KOR-ACC-79 Routing Protocol Flap SOP

**Node:** KOR-ACC-79 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4983

## Description

The occurrence of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Arista Access Node KOR-ACC-79 at Koramangala Datacenter can be attributed to the dynamic nature of routing protocols, which are designed to adapt to changing network topologies and failures. In this context, the Access role is particularly susceptible to routing protocol flaps due to its position as a gateway between the core network and the access layers. As a result, any instability or misconfiguration in the Access node's routing protocols can lead to a cascade of failures, causing the network to flap between different routing states. This can be exacerbated by the lack of adequate routing protocol stabilization mechanisms on the Access node, leading to an increased likelihood of routing protocol flaps.

## Pre-requisite Checks

- Check: Verify that the Koramangala Datacenter is accessible and power is present to the KOR-ACC-79 node.

- Check: Verify that the redundant links to the Access node are functioning correctly and there are no alarms on the redundant link.

- Check: Verify that the BGP, OSPF, and ISIS routing protocols are configured correctly and there are no misconfigurations.

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP route churn due to misconfigured BGP communities or inadequate BGP route reflection. | High |
| OSPF adjacency loss due to misconfigured OSPF interfaces or inadequate OSPF neighbor relationships. | Medium |
| ISIS LSP flooding due to misconfigured ISIS interfaces or inadequate ISIS level settings. | Low |
| Arista EOS software bug causing routing protocol instability. | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 1, local router ID is 192.0.2.1 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-rumor, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path, Description: This command provides an overview of the BGP routing table and can help identify any routing protocol instability or misconfigurations.
```

```
Command: show ospf neighbor, Expected Output: OSPFv2 Neighbor Table Neighbor ID Pri State Dead Time Address Interface 192.0.2.2 128 Full 00:00:30 192.0.2.2 Eth1/1, Description: This command provides an overview of the OSPF neighbor table and can help identify any OSPF adjacency loss or misconfigurations.

Command: show isis database, Expected Output: ISISv4 Database Level 2 Database 192.0.2.2/32, advertised by 192.0.2.1, metric 10, type explicit-null Level 1 Database 192.0.2.1/32, advertised by 192.0.2.2, metric 10, type explicit-null, Description: This command provides an overview of the ISIS database and can help identify any ISIS LSP flooding or misconfigurations.
```

## Resolution Steps

Step 1: Step: Verify and correct BGP routing protocol configuration., Command: show run | include bgp, Description: Verify that the BGP routing protocol is configured correctly and there are no misconfigurations. If any issues are found, correct the configuration and verify the changes.

Step 2: Step: Verify and correct OSPF routing protocol configuration., Command: show run | include ospf, Description: Verify that the OSPF routing protocol is configured correctly and there are no misconfigurations. If any issues are found, correct the configuration and verify the changes.

Step 3: Step: Verify and correct ISIS routing protocol configuration., Command: show run | include isis, Description: Verify that the ISIS routing protocol is configured correctly and there are no misconfigurations. If any issues are found, correct the configuration and verify the changes.

## Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 1, local router ID is 192.0.2.1 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-rumor, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path, Description: Verify that the BGP routing table is stable and there are no routing protocol instability or misconfigurations.

Command: show ospf neighbor, Expected Output: OSPFv2 Neighbor Table Neighbor ID Pri State Dead Time Address Interface 192.0.2.2 128 Full 00:00:30 192.0.2.2 Eth1/1, Description: Verify that the OSPF neighbor table is stable and there are no OSPF adjacency loss or misconfigurations.

Command: show isis database, Expected Output: ISISv4 Database Level 2 Database 192.0.2.2/32, advertised by 192.0.2.1, metric 10, type explicit-null Level 1 Database 192.0.2.1/32, advertised by 192.0.2.2, metric 10, type explicit-null, Description: Verify that the ISIS database is stable and there are no ISIS LSP flooding or misconfigurations.
```

## Roll-back Procedures

- Step: Restore the previous configuration., Command: copy run start, Description: Restore the previous configuration by copying the running configuration to the startup configuration.

- Step: Reapply the corrected configuration., Command: copy start run, Description: Reapply the corrected configuration by copying the startup configuration to the running configuration.